

Ex.No:3	ASSIGNMENT PROBLEM
Date:	

AIM:

To understand and solve the assignment problem using python ,assignment team of workers.

PROCEDURE:

1. **Input Data:** Define the cost matrix where each element represents the cost of assigning a particular task to a specific worker.
2. **Algorithm:** Use an optimization algorithm like the Hungarian algorithm (or the Munkres algorithm) to find the optimal assignment that minimizes the total cost.
3. **Output:** Display the optimal assignment along with the minimum total cost.

Problem-1:

JOB	J1	J2	J3	J4
W1	1	9	9	9
W2	9	1	9	9
W3	9	9	1	90
W4	9	9	9	1

Code:

```
import numpy as np
from scipy.optimize import linear_sum_assignment

cost_matrix =
    np.array([
        [1, 9, 9,
         9],
        [9, 1, 9, 9],
        [9, 9, 1, 9],
        [9, 9, 9, 1]
    ])

row_ind, col_ind = linear_sum_assignment(cost_matrix)

min_cost = cost_matrix[row_ind, col_ind].sum()

print("Optimal Assignments:", list(zip(row_ind, col_ind)))
print("Minimum Assignment Cost:", min_cost)
```

Output:

Optimal Assignments: [(0, 0), (1, 1), (2, 2), (3, 3)] Minimum
Assignment Cost: 4

Result :

Thus, The Assignment Problem to minimize the cost to assign a person in a job using python was executed successfully.